

day-23 Linked List

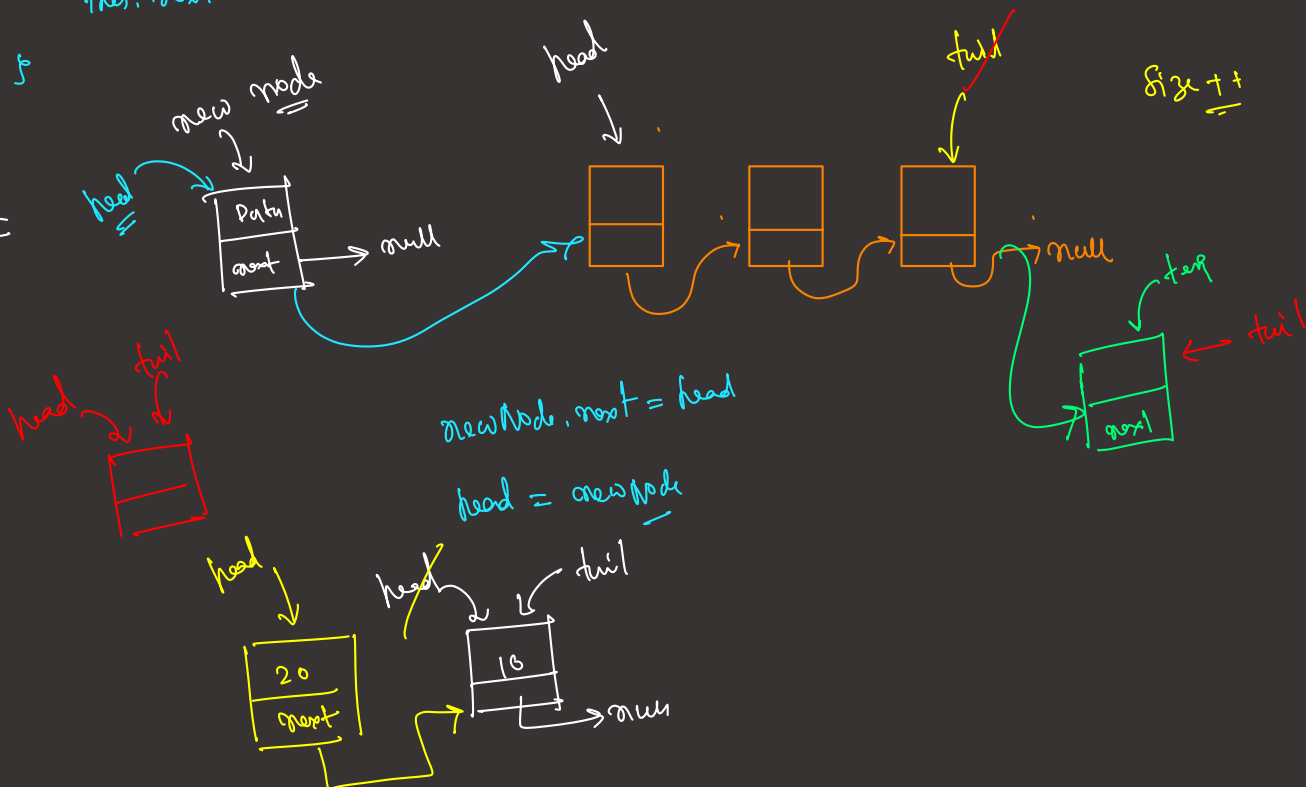
```
class Node {  
    int data;  
    Node next;
```

```
    public Node (int data) {  
        this.data = data;  
        this.next = null;
```

```
    }  
}
```

add first

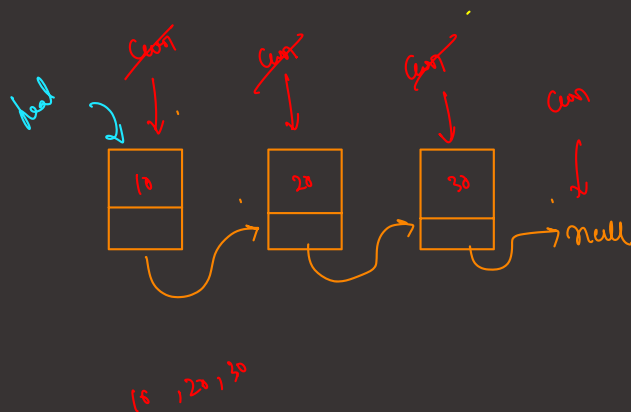
$O(1)$



$newNode.next = head$

$head = newNode$

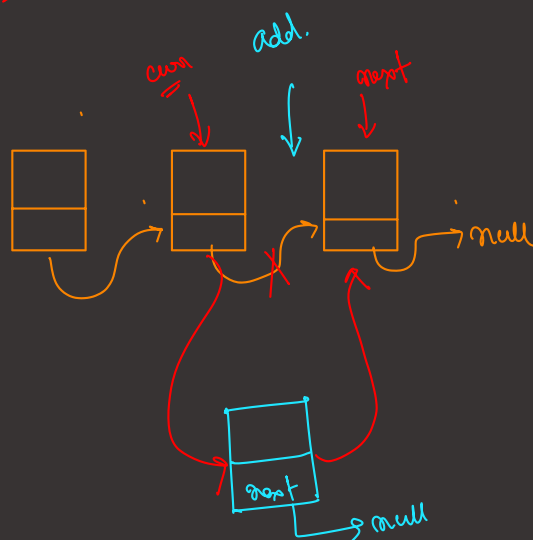
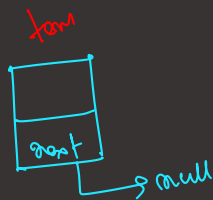
Print
 $O(n)$



add at any index

idx = 2

$O(n)$



curr, next = temp -
temp, next = next

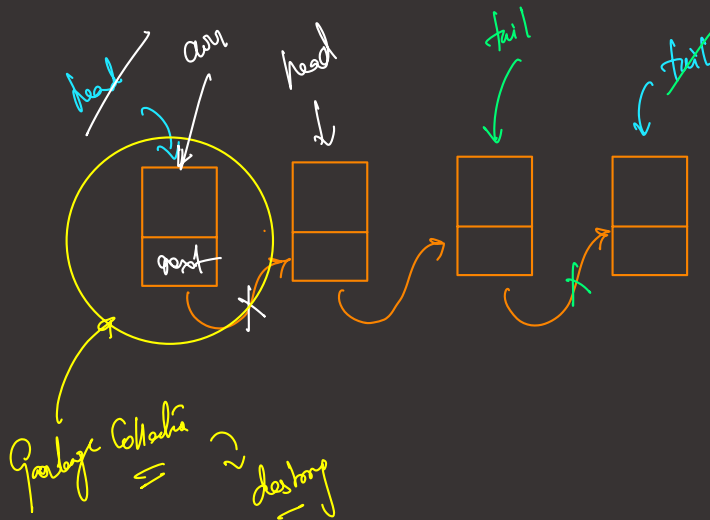
20 → 10 → 30

20 → 10 → 50 → 30

initially

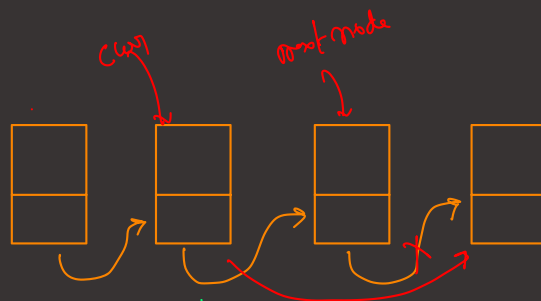
curr = head
next = head, next

Remove 0 position



Remove any idx

idx = 2



curr, next = nextNode, next;
nextNode = null;

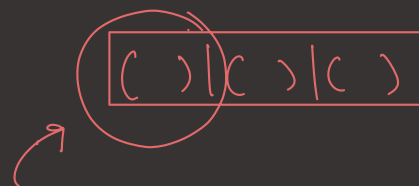
curr = head, nextNode = head, next;
for (int i = 0; i < !idx - 1; i++)
curr = curr, next;

H.W Search in a LL
delete a key.

Priority on for.

day-20
 class pair {
 int data;
 int freq;
 }

ArrayList < Pair > list = new ArrayList < > ();



Search

↙
while loop

→ Recursion

↖ return index
 Search (head, value, i) {

if (head == null) {
 return -1;

}

if (head.data == value) {
 return i;

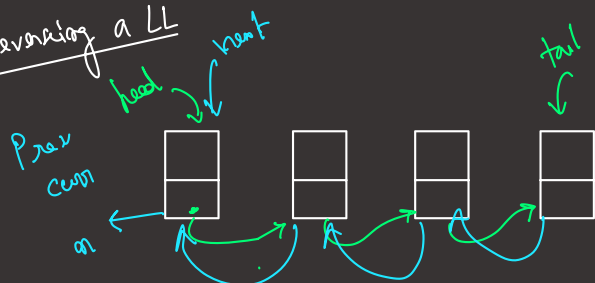
}

return Search (head.next, value, i+1);

}

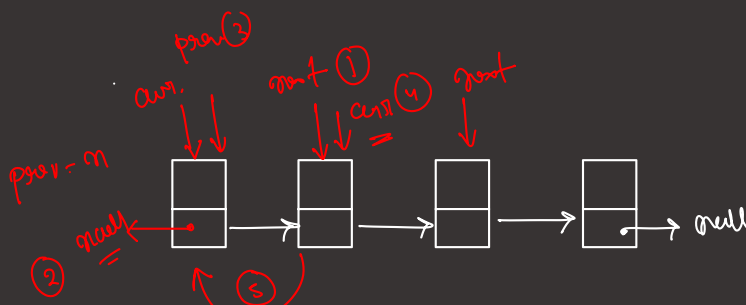
Reversing a LL

$\rightarrow O(n)$



prev = null
 curr = head
 next = null
 initial

next = curr.next
 curr = curr.next
 next.next = prev
 prev = curr



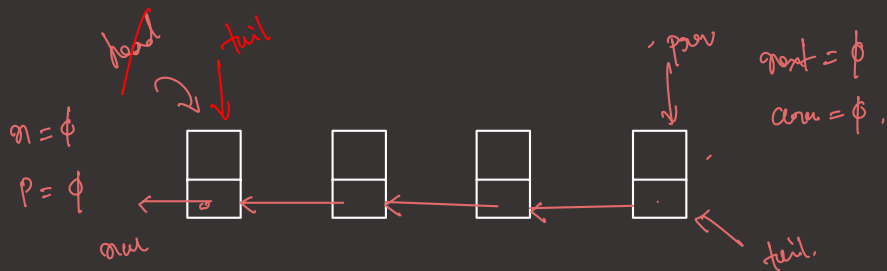
while (curr != null) {
 ① next = curr.next

② curr.next = prev

③ prev = curr

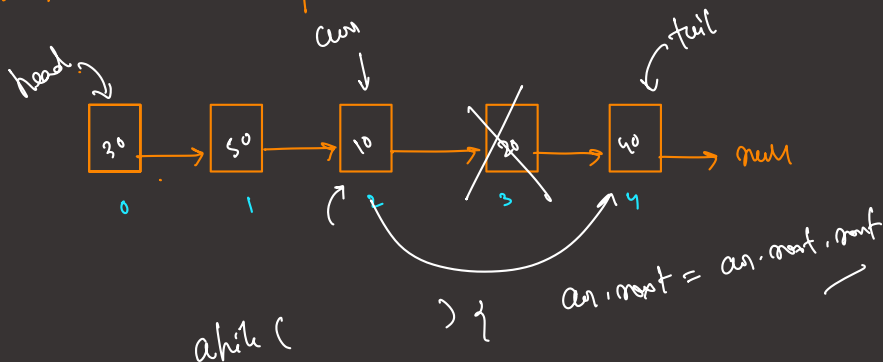
④ curr = next

initial
 prev = null
 curr = head
 next = null



tail = head
 head = prev

find and remove nth node from End

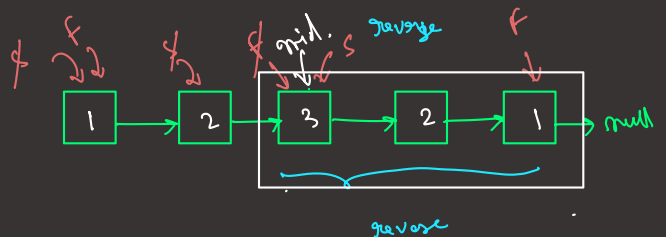
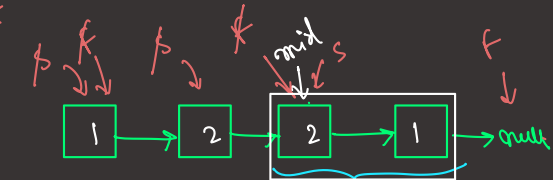


Size = 5
 I have to delete 3rd node
 Size - m
 $5 - 2 = 3$

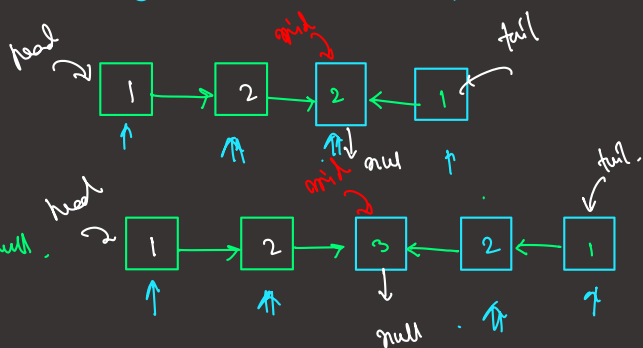
Q Check a linked list is palindrom or not

1 2 2 1 ✓ 1 2 3 2 1 ✓
 2 4 4 2 ✓
 2 4 4 3 ✗

Slow +1
 Fast +2



- to find mid node
- reverse the 2nd half



if (f == null or f->next == null)
 Slow → mid

